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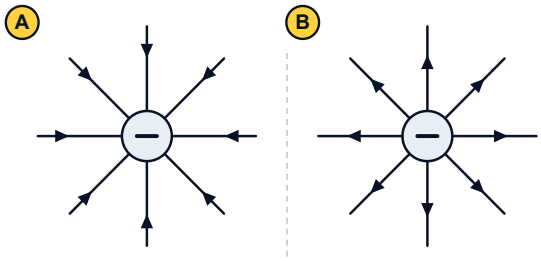
# Static charge and electric fields

**Electricity**
Paper 1, Separate Physics only
19 questions

Cover the answers. Answer each question out loud or on paper, then check. Tick the ones you know. Learn the 9 starred (★) questions first.

| | QUESTION | ANSWER | ✓ |
|---|---|---|---|
| 1 ★ | **Describe how insulating materials can become electrically charged.** | By rubbing them together: friction transfers electrons between them. | ☐ |
| 2 ★ | **Describe how a material can become negatively charged.** | By gaining electrons. | ☐ |
| 3 | **Describe how a material can become positively charged.** | By losing electrons. | ☐ |
| 4 ★ | **Describe what happens when two charged objects are brought close together.** | They exert a force on each other: like charges repel, opposite charges attract. | ☐ |
| 5 | **State whether electrostatic force is a contact force or non-contact force.** | Non-contact force | ☐ |
| 6 | **Explain why an insulator has no overall charge before it is rubbed.** | It has equal amounts of positive and negative charge. | ☐ |
| 7 | **Name the type of field that surrounds every charged object.** | An electric field | ☐ |
| 8 | **State where the electric field around a charged object is strongest.** | Closest to the object. | ☐ |
| 9 ★ | **Describe how the strength of an electric field changes with distance from the charge.** | The field gets stronger closer to the charge. | ☐ |
| 10 ★ | **State the direction in which electric field lines point.** | Away from positive charges and towards negative charges. | ☐ |

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| | QUESTION | ANSWER | ✓ |
|---|---|---|---|
| 11 | **Which diagram, A or B, shows the electric field around a negative charge? Explain.**<br>  | A.<br>The field lines point towards the negative charge. | ☐ |
| 12 | **Explain why a charged object experiences a force when close to another charged object.** | The electric field of each object exerts a force on the other object. | ☐ |
| 13 | **State how the size of the force between two charged objects can be increased.** | By moving them closer together. | ☐ |
| 14 | **State what happens when two objects with the same type of charge are brought together.** | They repel each other. | ☐ |
| 15 | **State what happens when two objects with opposite charges are brought together.** | They attract each other. | ☐ |
| 16 ★ | **A plastic rod becomes negatively charged when it is rubbed with a cloth. State and explain the charge on the cloth.** | Positive (equal in size): the cloth lost the electrons that the rod gained. | ☐ |
| 17 ★ | **What is an electric field?** | The region around a charged object where another charged object experiences a force. | ☐ |
| 18 ★ | **Describe the electric field around an isolated positively charged sphere.** | Straight lines pointing radially outwards, away from the sphere. | ☐ |
| 19 ★ | **Explain how a spark is produced between a charged object and an earthed conductor.** | The strong electric field ionises the air, so charge can flow across the gap. | ☐ |

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